An unbounded gap between minimum degree and the minor-connectivity ceiling

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Abstract

For a nonempty finite simple graph G, let

$$\kappa^*(G) = \max\{\kappa(H) : H \text{ is a nonempty minor of } G\},$$

where κ denotes vertex connectivity. We determine this parameter exactly on the independent blow-ups of the twelve-vertex Mader graph M_{12}. For every integer $t \geq 2$,

$$\kappa^*\big(M_{12}[\overline{K}_t]\big) = \left\lfloor \frac{9t}{2} \right\rfloor, \qquad \delta\big(M_{12}[\overline{K}_t]\big) = 5t.$$

Consequently, the difference between minimum degree and the largest connectivity of a minor is unbounded, giving infinitely many counterexamples to a conjecture of Barát. For context, we also prove the elementary identity $\kappa^*(K_s \vee G) = s + \kappa^*(G)$; applied to M_{12}, it already answers negatively the question whether sufficiently large minimum degree guarantees $\kappa^*(G) \geq \delta(G)$, but only with constant deficit. The independent blow-ups make that deficit unbounded. Their lower bound is supplied by an explicit uniform minor model. For the upper bound, a two-fibre separator localizes every relevant minor to a torso that is the join of three elementary graphs. Separating mixed and pure branch sets then reduces the problem to three sharp full-quotient profiles and an anticomplete-shores formula for connectivity. The proof is theoretical; accompanying computations only check the construction.

1 Introduction

All ambient graphs in this paper are nonempty, finite and simple; auxiliary graphs in decompositions may be empty. For a nonempty graph G, define the minor-monotone ceiling of vertex connectivity by

$$\kappa^*(G) = \max\{\kappa(H) : H \preccurlyeq G,\ |V(H)| \geq 1\}. \tag{1}$$

where $H \preccurlyeq G$ means that H is a minor of G. We use the convention $\kappa(K_n) = n - 1$. The parameter in (1) is denoted $\lceil \kappa \rceil$ in the minor-monotone ceiling literature.

Barát conjectured that every graph of minimum degree at least $k + 1$ contains a k-connected minor. Equivalently, the conjecture asserts

$$\kappa^*(G) \geq \delta(G) - 1 \tag{2}$$

for every nonempty graph [1]. Barioli, Barrett, Fallat, Hall, Hogben, Shader, van den Driessche and van der Holst studied minor-monotone ceilings and recorded a twelve-vertex graph M_{12} with

$$\delta(M_{12}) = 5, \qquad \kappa^*(M_{12}) = 4. \tag{3}$$

Thus M_{12} attains equality in (2), rather than contradicting it [2]. More recently, Barioli, Fallat, Gupta and Li used the same example when asking [3] whether there is a universal constant c such that

$$\kappa^*(G) \geq \delta(G) \tag{4}$$

whenever $\delta(G) \geq c$.

There is a simple first amplification of (3). Proposition 2 below shows that joining a graph with a clique increases its connectivity ceiling and minimum degree by the clique order. Thus $K_s \vee M_{12}$ has arbitrarily large minimum degree but retains deficit one, already answering (4) negatively. Our main result is stronger: independent blow-ups of M_{12} have an exact connectivity ceiling and a deficit that grows linearly.

Theorem 1. *For every integer $t \geq 2$, let*

$$G_t = M_{12}[\overline{K}_t].$$

Then

$$\kappa^*(G_t) = \left\lfloor \frac{9t}{2} \right\rfloor. \tag{5}$$

Since $\delta(G_t) = 5t$, Theorem 1 gives

$$\delta(G_t) - \kappa^*(G_t) = \left\lceil \frac{t}{2} \right\rceil. \tag{6}$$

In particular, (2) fails for every $t \geq 3$, and the difference in (6) is unbounded. This strengthens the fixed-deficit obstruction supplied by clique joins.

The proof has two complementary parts. The lower bound in (5) comes from a balanced family of explicit branch sets in the original graph. For the upper bound, the natural $2t$-vertex separator of G_t localizes any more than $2t$-connected minor to a $7t$-vertex torso. That torso has a three-factor join decomposition. Every branch set meeting two factors becomes universal in the quotient, while branch sets contained in one factor are controlled by simple full-quotient profiles. The resulting capacity inequalities give the matching upper bound.

2 A baseline amplification by clique joins

The ordinary connectivity formula for graph joins is classical. Barrett, Fallat, Hall and Hogben used it in their study of the connectivity ceiling [4]. For that ceiling it gives the following exact identity.

Proposition 2. *For every nonempty finite simple graph G and every integer $s \geq 0$,*

$$\kappa^*(K_s \vee G) = s + \kappa^*(G). \tag{J1}$$

Moreover,

$$\delta(K_s \vee G) = s + \delta(G). \tag{J2}$$

Proof. Choose a minor F of G with $\kappa(F) = \kappa^*(G)$. The graph $K_s \vee F$ is a minor of $K_s \vee G$, and the join-connectivity formula gives

$$\kappa(K_s \vee F) = s + \kappa(F).$$

This proves the lower bound in (J1).

For the reverse inequality, take a nonempty minor H of $K_s \vee G$ and a branch-set model for it. Retain every touching edge between branch sets, thereby obtaining a supergraph H^+ of H. Suppose that r branch sets meet the K_s side. Then $r \le s$, each of these branch sets is universal in H^+, and the remaining branch sets form a full branch-set quotient F in G. If F is nonempty, then

$$\kappa(H) \le \kappa(H^+) = r + \kappa(F) \le s + \kappa^*(G).$$

If F is empty, then $H^+ = K_r$ with $r \ge 1$, so $\kappa(H) \le r - 1 \le s + \kappa^*(G)$. Maximizing over H proves (J1).

For (J2), old vertices acquire s new neighbours, while each new clique vertex has degree $s - 1 + |V(G)|$. Since $\delta(G) \le |V(G)| - 1$, their minimum is $s + \delta(G)$. □

Applying Proposition 2 to (3) gives

$$\delta(K_s \vee M_{12}) = s + 5, \qquad \kappa^*(K_s \vee M_{12}) = s + 4. \tag{J3}$$

Thus the minimum degree in (J3) is unbounded while the deficit stays equal to one. This already disproves the existence of a threshold in (4). The remainder of the paper proves the stronger unbounded-deficit statement.

3 The graph M_{12} and its independent blow-ups

We give a labelled definition so that the construction does not depend on a drawing. Take two disjoint copies of $C_4 \vee K_2$. In the first copy, let the cycle vertices be x_0, x_1, x_2, x_3 in cyclic order and let a_x, b_x be the two adjacent vertices joined to the entire cycle. In the second copy use $y_0, y_1, y_2, y_3, a_y, b_y$. Add the six cross edges

$$x_0 y_i \quad \text{and} \quad y_0 x_i \qquad (i = 1, 2, 3). \tag{7}$$

The resulting graph is M_{12}. We use the numerical labels

$$\begin{aligned} 0, 1, 2, 3, 4, 5 &= x_0, x_1, x_2, x_3, a_x, b_x, \\ 6, 7, 8, 9, 10, 11 &= y_0, y_1, y_2, y_3, a_y, b_y. \end{aligned} \tag{8}$$

The graph has 12 vertices and 32 edges. Its vertices 0 and 6 have degree seven and all other vertices have degree five. The set $\{0, 6\}$ is a separator; after its deletion, the two components have vertex sets $\{1, 2, 3, 4, 5\}$ and $\{7, 8, 9, 10, 11\}$.

For an integer $t \geq 1$, the independent blow-up $G_t = M_{12}[\overline{K}_t]$ has vertex set $V(M_{12}) \times \{0, \ldots, t-1\}$. Distinct vertices (u, i) and (v, j) are adjacent precisely when $uv \in E(M_{12})$. Thus

$$|V(G_t)| = 12t, \qquad |E(G_t)| = 32t^2, \qquad \delta(G_t) = 5t. \tag{9}$$

The t vertices above one base vertex will be called a fibre. We write $I_m = \overline{K}_m$ for the independent graph on m vertices.

4 Two elementary connectivity tools

For a graph J, define

$$\beta(J) = \max(\{0\} \cup \{|X| + |Y| : X, Y \neq \varnothing,\ X \cap Y = \varnothing,\ E_J(X, Y) = \varnothing\}). \tag{10}$$

Thus $\beta(J) = 0$ for every complete graph, including the empty graph and the one-vertex graph. The next formula will only be used when J is not complete.

Lemma 3. *If J is a noncomplete graph on n vertices, then*

$$\kappa(J) = n - \beta(J). \tag{11}$$

Proof. If X and Y are nonempty anticomplete sets, deleting all other vertices disconnects J, so $\kappa(J) \leq n - |X| - |Y|$. Conversely, delete a minimum vertex cut and partition the components that remain into two nonempty unions X and Y. These sets are anticomplete and have total size $n - \kappa(J)$. Maximizing over X, Y gives (11). □

Recall that the join $J_1 \vee J_2$ is obtained from disjoint copies of J_1 and J_2 by adding every edge between them. In a join of several graphs, anticomplete shores must lie in a single factor. Consequently, if

$$J = K_u \vee J_1 \vee \cdots \vee J_m$$

is noncomplete, then

$$\kappa(J) = u + \sum_{i=1}^{m} |V(J_i)| - \max_i \beta(J_i). \tag{12}$$

Given pairwise disjoint connected branch sets in a graph F, their **full branch-set quotient** has one vertex per branch set and an edge for every pair of branch sets that touch in F. It is a minor of F obtained without a final edge-deletion step. The distinction matters: the following profiles are false for arbitrary minors after unrestricted edge deletion, but hold for the full quotients used in our upper-bound argument. For connected noncomplete graphs, the identity (11), in an equivalent complement-biclique form, also appears in Barrett, Fallat, Hall and Hogben [4].

We need three elementary full-quotient profiles.

Lemma 4. *Let $t, B \geq 1$ be integers, and let J_i be a full branch-set quotient of the indicated graph.*

1. *If J_1 comes from $K_t \sqcup I_t$, then either J_1 is complete and $|V(J_1)| \leq t$, or $\beta(J_1) = |V(J_1)|$.*
2. *If J_2 comes from I_{2t} and $\beta(J_2) \leq B$, then $|V(J_2)| \leq B$.*
3. *If J_3 comes from $K_t \sqcup K_{t,t}$, then one of the following holds: J_3 is empty; both components contribute vertices and $\beta(J_3) = |V(J_3)|$; only the K_t component contributes and J_3 is complete of order at most t; or only the $K_{t,t}$ component contributes and*

$$|V(J_3)| \leq \min\{2t, t + B\} \tag{13}$$

whenever $\beta(J_3) \leq B$.

Proof. A branch set cannot join different components of a graph. Because every touching edge is retained, a full quotient of $K_t \sqcup I_t$ is a clique together with isolated vertices. If it is not complete, its vertices can be partitioned into two nonempty anticomplete shores, proving the first assertion. The second assertion is immediate because a full quotient of an independent graph is independent.

For the third assertion, the empty case is immediate, and only the last case needs proof. We have

$$\operatorname{tw}(K_{t,t}) = t. \tag{14}$$

Taking one bipartition class together with each vertex of the other class gives a tree decomposition of width t. In the opposite direction, contracting $t-1$ disjoint cross edges and retaining the two unmatched vertices gives a K_{t+1} minor. Treewidth is minor-monotone, and vertex connectivity is at most treewidth. Hence every minor of $K_{t,t}$ has connectivity at most t, and every complete minor has order at most $t+1$.

If J_3 is noncomplete, Lemma 3 gives

$$|V(J_3)| - \beta(J_3) = \kappa(J_3) \leq t.$$

If J_3 is complete, then $|V(J_3)| \leq t + 1 \leq t + B$. Together with the trivial bound $|V(J_3)| \leq 2t$, this proves (13).

□

5 A uniform lower-bound model

Fix $t \geq 2$ and put

$$a = \left\lfloor \frac{t}{2} \right\rfloor, \qquad b = \left\lceil \frac{t}{2} \right\rceil. \tag{15}$$

Using the labels in (8), form the following t connected branch sets in G_t:

$$\begin{aligned} A_i &= \{(0,i),(6,i),(7,i),(10,i),(11,i)\} && (0 \leq i < a), \\ B_j &= \{(1,j),(6,a+j)\} && (0 \leq j < b). \end{aligned} \tag{16}$$

Add as singleton branch sets all unused vertices in the six fibres above $0,1,2,3,4,5$. There are t branch sets in (16) and $5t$ singletons, hence $6t$ branch sets in total.

The sets A_i are connected by the edges represented by (7) and the two copies of $C_4 \vee K_2$; each B_j is connected by the edge $1\,6$. The sets are pairwise disjoint. Moreover, every branch set in (16) is adjacent to every other branch set in the quotient. Thus the corresponding t quotient vertices are universal.

Let Q_t be the full branch-set quotient. In the complement of Q_t, the t universal vertices are isolated. The four nontrivial components have orders

$$t, \qquad t, \qquad 2t - a, \qquad 2t - b. \tag{17}$$

Each component is complete. The two larger components join the remaining vertices above 0 to the fibre above 2, and the remaining vertices above 1 to the fibre above 3; the other two components are the fibres above 4 and 5. Hence the largest complete bipartite subgraph in $\overline{Q_t}$ has total shore size

$$\max\{t, 2t - a, 2t - b\} = \left\lceil \frac{3t}{2} \right\rceil .$$

Equivalently, the maximum total size of two anticomplete shores in Q_t is $\lceil 3t/2 \rceil$. Lemma 3 now gives

$$\kappa(Q_t) = 6t - \left\lceil \frac{3t}{2} \right\rceil = \left\lfloor \frac{9t}{2} \right\rfloor . \tag{18}$$

Since Q_t is a minor of G_t,

$$\kappa^*(G_t) \geq \left\lfloor \frac{9t}{2} \right\rfloor . \tag{19}$$

6 Localization to a torso

Let S_t be the $2t$ vertices in the fibres above 0 and 6. The graph $G_t - S_t$ has two components of order $5t$. Fix the component above $\{1, 2, 3, 4, 5\}$ and call it C_t. Let T_t be obtained from $G_t[C_t \cup S_t]$ by completing S_t to a clique. Thus T_t has $7t$ vertices.

Lemma 5. *If G_t has an r-connected minor with $r > 2t$, then T_t has an r-connected minor.*

Proof. Let H be an r-connected minor of G_t, represented by pairwise disjoint connected branch sets. Let $X \subseteq V(H)$ index the branch sets that meet S_t. Since the branch sets are disjoint,

$$|X| \leq |S_t| = 2t < r.$$

The graph $H - X$ is connected and has at least two vertices. Every branch set indexed by $V(H) \setminus X$ lies in one component of $G_t - S_t$. As there are no edges between those two components, all these branch sets lie on the same side; by symmetry, take that side to be C_t.

Restrict each branch set meeting S_t to $C_t \cup S_t$. The restricted set is nonempty. Any excursion through the discarded component enters and leaves at vertices of S_t, so an added clique edge in $T_t[S_t]$ replaces that excursion. The same clique restores any adjacency between two restricted branch sets that was realized through the discarded side. Adjacencies to pure branch sets in C_t remain present. The restricted sets therefore form an H-minor model in T_t. $\square$

Since $\lfloor 9t/2 \rfloor + 1 > 2t$, Lemma 5 shows that the upper bound in Theorem 1 follows once we prove

$$\kappa^*(T_t) \le \left\lfloor \frac{9t}{2} \right\rfloor. \tag{20}$$

7 The three-factor torso

Partition the seven fibres of T_t into

$$F_1 = \{0, 2\}, \qquad F_2 = \{1, 3\}, \qquad F_3 = \{6, 4, 5\}.$$

The definition of M_{12} and the completion of S_t give

$$\begin{aligned} F_1 &\cong K_t \sqcup I_t, \\ F_2 &\cong I_{2t}, \\ F_3 &\cong K_t \sqcup K_{t,t}, \end{aligned}$$

and every vertex in one factor is adjacent to every vertex in either other factor. Therefore

$$T_t = (K_t \sqcup I_t) \vee I_{2t} \vee (K_t \sqcup K_{t,t}). \tag{21}$$

Consider an arbitrary minor model in T_t and retain every edge of its full branch-set quotient. Adding these edges cannot decrease vertex connectivity, so it suffices to bound the connectivity of the full quotient.

Call a branch set mixed if it meets at least two of the three factors in (21), and pure otherwise. Every mixed branch set becomes a universal quotient vertex. Indeed, given another branch set, one of the factors met by the mixed set differs from a factor met by the other set, and a join edge supplies the required adjacency. The same argument shows that two mixed branch sets are adjacent.

Let u be the number of mixed branch sets. For $i = 1, 2, 3$, the pure branch sets contained in F_i form a full branch-set quotient H_i of that factor, and in particular a minor of it; write $n_i = |V(H_i)|$. The full quotient is

$$H = K_u \vee H_1 \vee H_2 \vee H_3. \tag{22}$$

Each mixed branch set uses at least two vertices, and each pure branch set uses at least one. Since $|V(T_t)| = 7t$,

$$2u + n_1 + n_2 + n_3 \le 7t. \tag{23}$$

8 The general upper bound

We now prove (20). First suppose that the quotient H in (22) is not complete. Set

$$B = \max_{1 \le i \le 3} \beta(H_i), \qquad s = n_1 + n_2 + n_3.$$

Then $B \geq 1$, and (12), (22) and (23) give

$$\begin{aligned}\kappa(H) &= u + s - B \\ &\leq \frac{7t - s}{2} + s - B = \frac{7t}{2} + \frac{s}{2} - B.\end{aligned} \tag{24}$$

We distinguish the possible contributions to H_3.

Case 1: both components of F_3 contribute. Lemma 4 gives $B \geq n_3$. Hence

$$\kappa(H) \leq u + n_1 + n_2.$$

Every mixed branch set uses at least one vertex of $F_1 \cup F_2$, and each pure branch set counted by $n_1 + n_2$ uses a different vertex there. Since $|F_1 \cup F_2| = 4t$,

$$u + n_1 + n_2 \leq 4t \leq \left\lfloor \frac{9t}{2} \right\rfloor. \tag{25}$$

Case 2: only the K_t component of F_3 contributes, or H_3 is empty. Here $n_3 \leq t$. If H_1 is noncomplete, Lemma 4 gives $n_1, n_2 \leq B$, and therefore $s \leq t + 2B$. Equation (24) yields

$$\kappa(H) \leq 4t. \tag{26}$$

If H_1 is complete, then $n_1 \leq t$ and $n_2 \leq B$, so $s \leq 2t + B$. Hence

$$\kappa(H) \leq \frac{9t}{2} - \frac{B}{2} \leq \left\lfloor \frac{9t}{2} \right\rfloor. \tag{27}$$

The final inequality uses $B \geq 1$ and the integrality of $\kappa(H)$.

Case 3: only the $K_{t,t}$ component of F_3 contributes. Suppose first that H_1 is noncomplete. Lemma 4 gives

$$s \leq 2B + \min\{2t, t + B\}. \tag{28}$$

If $B \leq t$, then $s \leq t + 3B$, and (24) gives

$$\kappa(H) \leq 4t + \frac{B}{2} \leq \frac{9t}{2}.$$

If $B \geq t$, then $s \leq 2t + 2B$, and (24) again gives $\kappa(H) \leq 9t/2$.

It remains to suppose that H_1 is complete. Now

$$s \leq t + B + \min\{2t, t + B\}. \tag{29}$$

For $B \leq t$, this gives $s \leq 2t + 2B$ and $\kappa(H) \leq 9t/2$. For $B \geq t$, it gives $s \leq 3t + B$ and

$$\kappa(H) \leq 5t - \frac{B}{2} \leq \frac{9t}{2}.$$

In every subcase, integrality yields

$$\kappa(H) \le \left\lfloor \frac{9t}{2} \right\rfloor. \tag{30}$$

The three cases are mutually exclusive and exhaustive for a noncomplete quotient.

Finally suppose that H is complete. Every nonempty H_i must be complete. Lemma 4 gives

$$n_1 \le t, \qquad n_2 \le 1, \qquad n_3 \le t+1,$$

and thus $s \le 2t+2$. By (23),

$$|V(H)| = u + s \le \frac{7t+s}{2} \le \frac{9t+2}{2}.$$

Since $\kappa(H) = |V(H)| - 1$, the same parity check gives

$$\kappa(H) \le \left\lfloor \frac{9t}{2} \right\rfloor. \tag{31}$$

Equations (30) and (31) prove (20). Lemma 5 transfers this upper bound to G_t, and (19) supplies equality. This completes the proof of Theorem 1.

9 Consequences

Corollary 6. *There is no constant C such that*

$$\kappa^*(G) \ge \delta(G) - C$$

for every nonempty finite simple graph G.

Proof. For G_t, the deficit is $\lceil t/2 \rceil$ by (6), and hence is unbounded.

□

Corollary 7. *There is no constant c such that every nonempty graph G with $\delta(G) \ge c$ satisfies $\kappa^*(G) \ge \delta(G)$.*

Proof. This already follows from (J3). It also follows from the stronger family G_t: the minimum degrees $\delta(G_t) = 5t$ are unbounded, while (6) is positive for every $t \ge 2$. □

For $t \ge 3$, let $k = 5t - 1$. Then G_t has minimum degree $k+1$ but

$$\kappa^*(G_t) = \left\lfloor \frac{9t}{2} \right\rfloor < 5t - 1 = k.$$

Thus the family gives infinitely many counterexamples to Barát's conjecture. The first member covered by this strict inequality is the 36-vertex graph G_3, for which $\delta(G_3) = 15$ and $\kappa^*(G_3) = 13$.

The formula also gives $\kappa^*(G_2) = 9$ and $\kappa^*(G_4) = 18$. The previously known value $\kappa^*(M_{12}) = 4$ is consistent with the same rounded expression at $t = 1$, although Theorem 1 is stated only for $t \ge 2$.

10 Reproducibility and evidence boundary

The proof of Theorem 1 is independent of exhaustive computation. The accompanying verification program reconstructs the branch sets in (16), checks that they are nonempty, disjoint and connected in the original graph, builds their full quotient, and computes its vertex connectivity. From the repository root, run

```
.\.venv\Scripts\python.exe research\M12-Minor-Connectivity-Gap\verification\
verify_general_lower_model.py
```

The frozen output covers $t = 2, \ldots, 8$ and reports the connectivities

$$9, 13, 18, 22, 27, 31, 36.$$

This finite calculation checks the stated construction and its implementation; it is not used to infer the universal quantifier in Theorem 1. The upper bound is entirely theoretical.

11 Prior work and novelty boundary

Barát's thesis records the conjecture leading to (2) [1]. The graph M_{12} originates in Mader's 1968 work [5]; Barioli et al. later introduced the relevant minor-monotone ceiling notation and established (3) for this graph [2]. Barrett, Fallat, Hall and Hogben developed nearby connectivity tools [4], including an equivalent form of (11) and the ordinary join-connectivity formula. Fijavž and Wood studied broader relations between graph minors and minimum degree [6]. Barioli, Fallat, Gupta and Li subsequently used M_{12} [3] to show that $\kappa^*(G) \geq \delta(G)$ does not hold for every graph and asked whether it holds above a universal minimum-degree threshold; Proposition 2 gives a negative answer, while Theorem 1 supplies the new independent-blow-up formula, makes the deficit unbounded, and yields infinitely many counterexamples to Barát's conjecture.

A targeted search did not locate the exact formula (5) or this independent blow-up family in prior work. This is a candidate novelty statement rather than proof of global priority. No claim of first discovery is made, and the historical assessment remains subject to specialist review and searches of sources not indexed in the databases consulted.

AI-assisted research and writing disclosure

Carptopus is the sole author and is responsible for the mathematical claims, source selection, revisions and public version of this manuscript. OpenAI Codex was used as an AI-assisted tool for literature discovery, proof exploration, exact verification, adversarial auditing and manuscript drafting.

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